



DEPARTMENT OF COMPUTER SCIENCE

BREWPOINT ISLAMABAD

FULL STACK COFFEE PICKUP ORDERING

PLATFORM

PROJECT REPORT

Cover Page Details

University: __AIR__

Course: Web Technologies

Semester Project

Project Title: BrewPoint Islamabad

Team Members & Roll Numbers: HANIYA NOOR (2502087) SANA AZIZ (2502083)

Table of Contents

(Update automatically in MS Word using References → Table of Contents)

Abstract

BrewPoint Islamabad is a full-stack coffee pickup ordering platform developed to streamline coffee ordering and pickup operations across four BrewPoint locations in Islamabad.

Introduction

Traditional coffee ordering often results in queues, delays, and inefficient order management. BrewPoint Islamabad digitizes the process through a modern web platform.

Problem Statement

Customers experience long waiting times and limited ordering flexibility. Business owners face challenges in managing orders and customer data efficiently.

Objectives

Provide online ordering, secure authentication, order tracking, pickup location selection, and administrative management.

Project Scope

The system supports customers, administrators, coffee catalog management, order management, reviews, and location management.

Technology Stack

Frontend: React, TypeScript, Tailwind CSS, Framer Motion, shadcn/ui.

Backend: ASP.NET Core Web API.

Database: SQL Server.

System Architecture

Presentation Layer (React) → API Layer (ASP.NET Core) → Data Layer (SQL Server).

Architecture Diagram Placeholder:

[Insert Architecture Diagram Here]

Frontend Design

The frontend includes Home, Authentication, Catalog, Coffee Details, Dashboard, Profile, Locations, and Admin Panel pages.

Backend Design

The backend uses ASP.NET Core Web API, Entity Framework Core, JWT Authentication, Authorization, DTOs, Validation, and Swagger.

Database Design

Entities include Users, Coffees, Orders, OrderItems, Reviews, and Locations.

ER Diagram Placeholder:

[Insert Database Diagram Here]

Database Tables

Users, Coffees, Orders, OrderItems, Locations, Reviews with proper relationships and foreign keys.

Authentication Module

JWT-based authentication with Signup, Login, Token Validation, and Role-Based Authorization.

User Features

Browse coffees, customize drinks, select pickup location, place orders, manage profile, and view order history.

Admin Features

Manage coffees, users, locations, reviews, and order statuses using CRUD operations.

Live Cup Builder

Implemented using Vanilla JavaScript DOM Manipulation with event listeners and real-time UI updates.

API Documentation

Authentication APIs, Coffee APIs, Order APIs, Location APIs, and Review APIs.

API Endpoints

POST /api/auth/signup
POST /api/auth/login
GET /api/auth/me
GET /api/coffees
GET /api/coffees/{id}
POST /api/orders
GET /api/orders/my-orders
GET /api/locations

Security Features

JWT authentication, password hashing, validation, CORS, role authorization, and centralized error handling.

Testing

Unit testing, API testing through Swagger/Postman, UI testing, and database validation.

Deployment

Frontend deployment on Vercel/Netlify, Backend deployment on Azure/Render/Railway, SQL Server hosting.

GitHub Repository Structure

frontend/
backend/
database/
report/
demo/

Project Screenshots

Home Page Screenshot Placeholder
Login Screenshot Placeholder
Signup Screenshot Placeholder
Catalog Screenshot Placeholder
Dashboard Screenshot Placeholder
Admin Panel Screenshot Placeholder

BrewPoint Islamabad

HomeCatalogLocationsDashboard

LoginSignup

JOIN BREWPOINT

Create your pickup account.

Full name

Phone

Email

Password

Signup

BrewPoint Islamabad

HomeCatalogLocationsDashboard

hano Logout

MEMBER COUNTER

Welcome back to the bar.

Track pickups, repeat favorites, and manage your BrewPoint ritual.

SIGN IN

Login

Demo: admin@brewpoint.com / Admin@123 or user@brewpoint.com / User@123.

Email

Password

Login

New here? [Create account](#)

CATALOG

Brewed signatures

**Classic Americano**

Rs 420

BLACK

Bold espresso stretched with hot water.

[Customize](#)**Dark Chocolate Mocha**

Rs 730

MOCHA

Deep cocoa, espresso, and steamed milk.

[Customize](#)**Espresso Shot**

Rs 340

ESPRESSO

A compact, high-clarity espresso.

[Customize](#)**Hazelnut Macchiato**

Rs 690

MACCHIATO

Layered espresso with hazelnut sweetness.

[Customize](#)**Honey Cinnamon Flat White**

Rs 700

FLAT WHITE

Flat white with honey and cinnamon.

[Customize](#)**Islamabad Iced Latte**

Rs 640

ICED

Cold espresso, milk, and a clean city break finish.

[Customize](#)**Midnight Black Brew**

Rs 520

BREWED

Dark, clean, late-eight filter coffee.

[Customize](#)**Pink Cloud Mocha**

Rs 700

MOCHA

Silky mocha with pink vanilla foam.

[Customize](#)**Rose Pistachio Latte**

Rs 780

**Saffron Cappuccino**

Rs 720

HOT

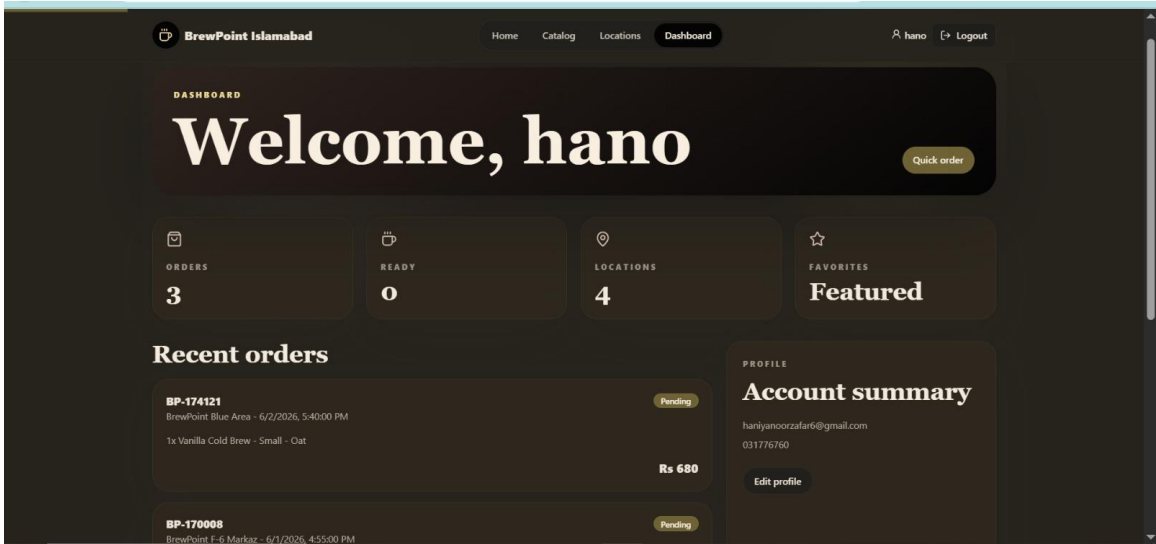
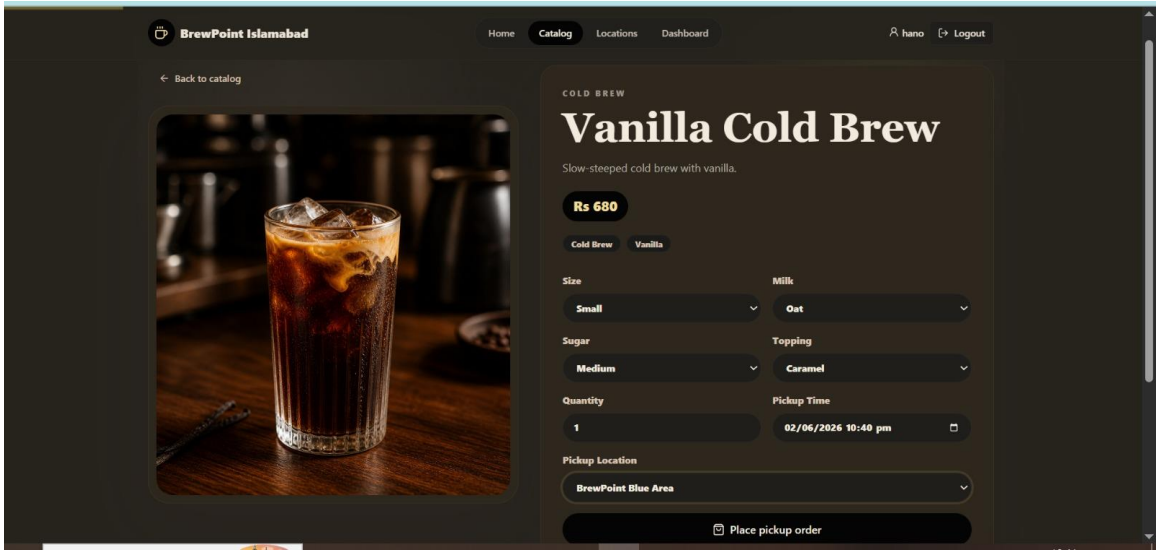
**Salted Caramel Latte**

Rs 740

**Vanilla Cold Brew**

Rs 680

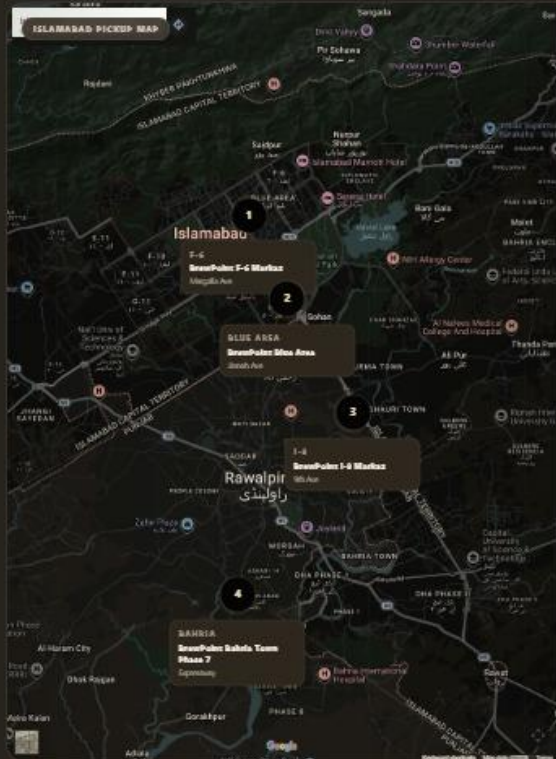
COLD BREW



PICKUP POINTS

BrewPoint across Islamabad

ISLAMABAD PICKUP MAP



MARKED LOCATIONS

Choose your counter.

- BrewPoint F-6 Markaz**
F-6 - MARGALLA AVE
Street 10, F-6 Markaz, Islamabad
+92 51 111 276 081
[Open map](#)
- BrewPoint Blue Area**
BLUE AREA - JINNAN AVE
Jinnah Avenue, Blue Area, Islamabad
+92 51 111 276 082
[Open map](#)
- BrewPoint I-8 Markaz**
I-8 - 8TH AVE
I-8 Markaz, Islamabad
+92 51 111 276 083
[Open map](#)
- BrewPoint Bahria Town Phase 7**
BAHRIA - EXPRESSWAY
Bahria Town Phase 7, Islamabad
+92 51 111 276 084
[Open map](#)

Please use appropriate visual markers for the four needed BrewPoint branches. The Open map buttons search each branch in Google Maps.

**BrewPoint F-6 Markaz**

Street 10, F-6 Markaz, Islamabad
+92 51 111 276 081
7:00 AM - 11:00 PM

**BrewPoint Blue Area**

Jinnah Avenue, Blue Area, Islamabad
+92 51 111 276 082
7:00 AM - 12:00 AM

**BrewPoint I-8 Markaz**

I-8 Markaz, Islamabad
+92 51 111 276 083
8:00 AM - 11:00 PM

**BrewPoint Bahria Town Phase 7**

Bahria Town Phase 7, Islamabad
+92 51 111 276 084
8:00 AM - 9:00 AM

 **BrewPoint Islamabad**

[Home](#) [Catalog](#) [Locations](#) [Dashboard](#)

[hano](#) [Logout](#)

CURRENT FAVORITES

A warmer menu, less noise.

Every card is pulled from the backend catalog and styled like a boutique roastery shelf: textured, grounded, and focused on the drink.

 Featured



Islamabad Iced Latte **Rs 640**

ICED

Cold espresso, milk, and a clean city-break finish.

[Customize →](#)

 Featured



Midnight Black Brew **Rs 520**

BREWED

Dark, clean, late-night filter coffee.

[Customize →](#)

 Featured



Pink Cloud Mocha **Rs 760**

MOCHA

Silky mocha with pink vanilla foam.

[Customize →](#)

 Featured



Rose Pistachio Latte **Rs 780**

LATTE

A fragrant latte with rose and pistachio.

[Customize →](#)

CUSTOM CUP

Live Cup Builder

This preview is powered by vanilla JavaScript DOM listeners inside React.

Size

Small

Milk

Oat

Sweet

Low

Top

None



Small cup - Oat milk - Low sweetness

 **BrewPoint Islamabad**

[Home](#) [Catalog](#) [Locations](#) [Dashboard](#)

[hano](#) [Logout](#)



HOW PICKUP WORKS

From craving to counter.

 01

Choose

Pick from 12 brewed signatures.

 02

Customize

Size, milk, sugar, toppings.

 03

Select

Choose your Islamabad pickup point.

 04

Collect

Track status until ready.

Future Enhancements

Online payment integration, mobile app support, loyalty rewards, AI recommendations, and real-time notifications.

Conclusion

BrewPoint Islamabad successfully demonstrates a modern full-stack web solution with responsive design, secure backend services, and efficient database management.

References

React Documentation, ASP.NET Core Documentation, SQL Server Documentation, Tailwind CSS Documentation, Framer Motion Documentation.

1. Home Page (Landing Page)

Purpose

The Home Page is the main entry point of BrewPoint Islamabad. It introduces users to the platform, highlights featured coffees, pickup locations, and provides quick navigation to important sections.

Main Components

Hero Section

- Large promotional banner
- Attractive coffee-themed design
- Call-to-action buttons
- Framer Motion animations

Featured Coffees

- Displays popular coffee products
- Coffee cards with images and pricing
- Direct links to product details

Coffee Carousel

- Interactive scrolling coffee showcase
- Highlights premium products

Live Cup Builder

- Built using Vanilla JavaScript DOM Manipulation
- Users customize coffee options
- Real-time visual updates without page reload

Pickup Locations Preview

- Displays BrewPoint branches
- Quick information about pickup points

Map Section

- Visual representation of locations
- Helps customers find nearest branch

Benefits Section

- Fast pickup service
- Premium coffee quality
- Convenient ordering process

2. Login Page

Purpose

Allows registered users to access their accounts securely.

Main Components

Login Form

- Email field
- Password field

Validation

- React Hook Form
- Zod validation

Authentication

- JWT-based authentication
- Secure login process

Navigation Links

- Link to Signup Page
- Redirect to Dashboard after successful login

3. Signup Page

Purpose

Allows new customers to create an account.

Main Components

Registration Form

- Full Name
- Email Address
- Phone Number
- Password

Form Validation

- Required field validation
- Email format validation
- Password strength validation

Account Creation

- Sends data to ASP.NET Core API
- Creates user record in SQL Server database

Dashboard Redirect

- Automatically redirects after successful registration

4. Catalog Page

Purpose

Displays all available coffee products.

Main Components

Catalog Header

- Attractive coffee menu banner

Search Functionality

- Real-time product searching
- Filters products by name and category

Coffee Cards

Each card contains:

- Coffee Image
- Coffee Name
- Category
- Price
- View Details Button

Dynamic Data Loading

- Data fetched from backend API
- No hardcoded products

5. Coffee Detail Page

Purpose

Provides detailed information and ordering functionality for a selected coffee.

Main Components

Product Information

- Coffee Image
- Name
- Description
- Price

Customization Options

Size Selection

- Small
- Medium
- Large

Milk Type

- Whole Milk
- Almond Milk
- Oat Milk

Sugar Level

- No Sugar
- Medium Sugar
- Extra Sugar

Toppings

- Chocolate
- Caramel
- Whipped Cream

Quantity Selection

- Customer chooses quantity

Pickup Location Selection

- Select one of four BrewPoint branches

Pickup Time

- User chooses collection time

Order Submission

- Sends order to backend API

6. Dashboard Page

Purpose

Provides personalized information for logged-in users.

Main Components

Welcome Section

- Displays customer name
- Personalized greeting

Quick Order Button

- Direct access to catalog

Order History

Displays:

- Previous Orders
- Order Status
- Pickup Information

Statistics Cards

May show:

- Total Orders

- Favorite Coffees
- Reviews
- Locations Visited

7. Profile Page

Purpose

Allows users to manage their personal information.

Main Components

User Information Form

- Full Name
- Phone Number

Profile Update

- Modify account information
- Save changes to database

Account Management

- View personal account details

8. Locations Page

Purpose

Displays all BrewPoint pickup locations in Islamabad.

Main Components

Location Header

- Branch overview section

Interactive Map

Shows:

- Branch locations
- Pickup points

Location Cards

Each card contains:

- Branch Name
- Address
- Contact Information
- Operating Hours

Branch Selection Support

Helps customers choose nearest pickup point

9. Admin Panel

Purpose

Allows administrators to manage the entire system.

Main Components

Coffee Management (CRUD)

Create

Add new coffee products

Read

View all products

Update

Modify coffee details

Delete

Remove products

User Management (CRUD)

View Users

Display all registered customers

Manage Accounts

Update or remove user records

Location Management (CRUD)

Add Locations

Create new branches

Edit Locations

Update branch information

Delete Locations

Remove branches

Order Management

View Orders

Monitor all customer orders

Update Status

- Pending
- Preparing
- Ready for Pickup
- Completed

Review Management

View Reviews

Monitor customer feedback

Moderate Reviews

Remove inappropriate reviews

Live Cup Builder (Special JavaScript Feature)

Purpose

Demonstrates Vanilla JavaScript and DOM Manipulation as required by the project.

Features

Event Listeners

Detects user selections.

Dynamic Updates

Updates cup preview instantly.

Real-Time Rendering

Changes appear without refreshing page.

Interactive User Experience

Provides immediate visual feedback.